



Review Article

Dietary management of dyslipidemia

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ABSTRACT

The rising burden of cardiovascular disease (CVD) has made the achievement of optimal lipoprotein levels a major public health priority. As nearly a fifth of global mortality is associated with dietary factors, and recommendations have been mired in controversy, a fresh look on the available data is attempted. Well established concepts regarding nutrition and cardiometabolic health, role of macronutrients, calories, and controversial foods are discussed followed by recommendations in the Indian context. A healthy dietary pattern rather than individual foods or nutrients is emphasized, and this is generally plant based with optional consumption of dairy, eggs, and meats within the suggested limits. Suggestions/recommendations are given for consumption of individual foods, remembering that choosing appropriate replacement foods is as important as restricting unhealthy foods.

1. Introduction

Dyslipidemia is a major risk factor for atherosclerotic cardiovascular disease (ASCVD),^{1,2} which accounts for nearly a third of total mortality, making the achievement of optimal lipoprotein levels a priority for community health.³ India experienced a near doubling of CVD mortality between 1990 and 2016 with a recent analysis ascribing 56.4 % of CVD risk to dietary factors, commensurate with the approximately 11 million deaths that are attributable to diet associated risk factors globally.^{3,4}

Studies on food items are subject to bias and uncertainty as these data are derived from small experimental studies and large observational studies, usually based on food frequency questionnaires (FFQ) filled by a small subset of the population being studied, typically at one point in time, and for a limited period.⁵ Furthermore, dietary patterns change over time, both as the population grows older and as new concepts become public, making diets difficult to control in the long-term in a free-living population. Inconsistent data and conflicting, often unsupported dietary advice over the last half century have led to confusion regarding a healthy diet.

Some concepts that are well established are summarized here:

1. Diet and exercise strongly influence cardio-metabolic risk factors. Easily assessed parameters like body weight, body fat (especially visceral fat), blood sugar, lipid levels, blood pressure, hepatic and renal function, lean muscle and bone mass, and more complex interactions like inflammation, oxidative stress, endothelial health,

and the gut microbiome are regulated and modified by our food and physical activity, probably through hormonal and other influences.⁶

2. Total macronutrients (carbohydrates, fats and proteins) are neutral for obesity, diabetes and CVD, implying that individual food choices rather than proportion of nutrients influence health.^{6,7} Large studies failed to show any major benefits with low fat or low saturated fat diets.^{8–14} Conversely, better outcomes were reported with higher fat diets^{15,16} providing more than 40 % of daily calories from fats without altering cholesterol levels, suggesting that the previous focus on total calories, macronutrients, total and saturated fats needs to be modified to look at dietary patterns, both for optimal lipoprotein levels and reducing CVD risk.^{6,7,10,12,13} Of the evaluated diets, the Mediterranean^{6,10,15,16} and Dietary Approaches to Stop Hypertension (DASH) diets have the maximum supportive evidence.^{6,10,17,18}
3. There is a strong link between weight gain, glucose intolerance and cardiometabolic disease; likewise, ≈15 kg weight loss by lifestyle changes or bariatric surgery can result in remission of type 2 diabetes (T2D) in ~80 % subjects.^{19,20} Maintaining ideal body weight is thus integral to maintaining insulin sensitivity, achieving optimal lipoprotein levels, and minimizing ASCVD risk.^{6,20}
4. Mounting evidence of a body weight set point in the hypothalamus suggests that the belief that obesity and weight gain are only due to energy imbalance (“calories in, calories out,” hypothesis) is an oversimplification of a complex metabolic process.^{6,21,22} Calorie restriction or exercise result in initial weight loss that is notoriously difficult to maintain. Weight maintenance is best achieved using healthy dietary patterns combined with regular exercise, without

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necessarily counting calories or relative proportions of macronutrients.^{6,21–26} Healthy foods result in greater satiety and/or have higher metabolic expenditure even if they are calorie-dense, while unhealthy foods are usually associated with quick bursts of energy followed by troughs, leading to craving for foods/snacks (related to brain reward mechanisms) predisposing to insulin resistance, obesity and potential adverse effects upon the gut microbiome.^{6,10,22,24,27–30}

5. In obese and/or insulin resistant (IR) individuals, the liver responds to excess calories (particularly refined carbohydrates) by increased secretion of triglyceride (TG)-rich very low density lipoprotein (VLDL) particles.³¹ Cholesteryl ester transfer protein (CETP) mediates the transfer of cholesterol and TGs between high density lipoproteins (HDL), LDL and VLDL, generating smaller cholesterol poor HDL and LDL particles.³² The smaller HDL particles are easily excreted by the kidneys, lowering HDL-C, whereas smaller LDL particles penetrate endothelium more easily and are vulnerable to oxidation, making them more atherogenic. The increased VLDL particles are more atherogenic (cholesterol-rich) and increase the Apo B particle number (Apolipoprotein B100 is the primary protein constituent of lipoproteins LDL and VLDL-their particle number serves as a better marker of atherogenicity than simply LDL-C levels), while calculated LDL cholesterol (LDL-C) remains around the normal range.^{1,31} Thus in individuals with metabolic syndrome, type 2 diabetes and hypertriglyceridemia, Apo B correlates better with ASCVD risk due to discordance between LDL particle number and calculated levels, the driver of atherogenic dyslipidaemia being excess calories, especially as refined carbohydrates.^{1,12,31} This is increasingly important as 1/3–1/2 of the adult population in many countries has diabetes or prediabetes,³³ and VLDL/remnant cholesterol contributes significantly to the burden of coronary artery disease.³⁴

2. Fats

Whereas polyunsaturated fatty acids (PUFA) appear to be the healthiest nutrient metabolically, the Food and Nutrition Board Institute of Medicine sets **No Tolerable Upper Intake Level for total fat**.³⁵ The intervention arms in the major trial supporting Mediterranean diet consumed 41 % calories from fats, which contributed to the removal of cut-offs for total fats in guidelines.³⁶ Only trans-fats (bakery items, fried foods, hydrogenated vegetable oils, most over-the-counter snacks) are unequivocally linked to increased ASCVD and hence are proscribed completely.^{6,10,11}

3. Total fats, saturated fats, cholesterol and cooking oils

Experimental studies carrying out direct substitution of nutrients,³⁷ as well as large cohort studies that model the effects of nutrient substitution uniformly conclude that iso-calorically replacing saturated fatty acids (SFA) or carbohydrates by PUFA is beneficial and by mono-unsaturated fatty acids (MUFA), neutral for the lipoproteins, whereas substituting SFA with total carbohydrates worsens TGs and HDL-C and lowers LDL-C, a change that is better or worse depending upon the quality of carbohydrate.^{6,10,12,37} However, foods are different from drugs, i.e. extrapolating the benefits of lower LDL-C with food does not guarantee similar CVD reduction as achieved with statins. Examples include failed trials with niacin,³⁸ CETP inhibitors³⁹ and estrogen supplements⁹; furthermore, benefits observed with foods have never been replicated with nutrients given as drugs.^{6,7}

Saturated fats have been considered a major risk factor for CVD^{6,10,11,40–42} owing to increased LDL-C levels, a change which may be offset by their other effects: increased HDL-C (maintaining the TC/HDL-C ratio) while substantially reducing triglycerides,³⁷ no major impact on Apo B,¹² and lower lipoprotein(a) levels.⁴³ Conclusive evidence of CVD risk is unproven: all 10 meta-analyses of observational studies evaluating the association of SFA with CVD were negative (only 2 reported benefits on replacing SFA with PUFA), as were 7 of 10

meta-analyses of RCTs; the 3 positive meta-analyses reported lower combined CVD endpoints, and not mortality.^{44–54} The positive analyses included the “4 core trials” that form the basis of AHA guidelines, which have been criticized for design flaws (poorly controlled), being outdated (over 50 years old) and comparing consumption of trans- and saturated fats (rather than SFA alone) vs PUFA.^{40,44} Finally, all 8 meta-analyses evaluating the association of foods rich in saturated fats with CVD were negative⁴⁴; a large trial reported lack of change in lipoproteins with higher SFA consumption in the setting of a natural diet with low refined carbohydrates.⁵⁵ There is even a study demonstrating least progression of coronary atherosclerosis in women who consumed more saturated fats than those consuming higher carbohydrates or PUFA.⁵⁶

PREDIMED reported 30 % reduction in CVD endpoints with Mediterranean diet, but had similar 9 % SFA intake in all arms, the difference being extra unsaturated fat intake in the two active arms in the context of a higher than usual fat intake (approximately 41 % of daily calories).^{15,36} Thus, some guidelines recommend that replacing refined carbohydrates with saturated fat could be metabolically healthy^{10,46,57}; an alternative suggestion is to limit total carbohydrate and SFA rather than SFA alone.¹⁰

One reason for the continuing controversy could be the reported association of plasma SFA levels with metabolic syndrome, diabetes and CVD, but it is noteworthy that these levels are proportional to the total carbohydrate, and not fat intake, as higher saturated fats with low carbohydrate consumption results in lower circulating SFA levels (due to greater fat oxidation and suppression of hepatic lipogenesis).^{6,10,45} Furthermore, SFA are heterogeneous, found in both healthy and unhealthy foods (for instance, cheese, curd, butter, chicken, seafood, meats, fried foods, oils and nuts), making judging a fat by its fatty acid content imperfect.^{6,10,45,58}

Randomized controlled trials of PUFA substitution or supplementation are inconsistent: Some substitution studies report reduction of CVD endpoints,^{10,35,59} but the results of the two largest studies^{60,61} reported reduction in cholesterol levels but a possible increase in mortality; hence, the results of meta-analyses depend upon the studies included or excluded. Reassuring data in favor of linoleic acid (n-6) comes from a recent large meta-analysis of prospective cohort studies that reported significant reductions in CVD, cancer and mortality.⁶²

The same is true of supplementation: whereas PREDIMED, which added fats as foods (nuts or extra-virgin olive oil [EVOO]) is positive,¹⁵ all trials of nutrient supplementation are negative, except REDUCE-IT,⁶³ which reported significant 20 % mortality reduction with supplemental n-3 fatty acid icosapent ethyl (purified ethyl ester of eicosapentaenoic acid, EPA) in subjects with elevated TGs on statins. As STatin Residual Risk Reduction With EpaNova in HiGH CV Risk PatientS With Hypertriglyceridemia (STRENGTH)⁶⁴ studying EPA + DHA combination and VITAL⁶⁵ testing vitamin D₃ and lower dose EPA + DHA combination reported neutral results, we are without clear answers.

In sum, healthy fat containing foods (e.g., nuts, EVOO, some dairy fats) may be beneficial in comparison to refined, processed carbohydrates, unsaturated fats are a healthy addition to diet replacing refined carbohydrates and unhealthy saturated fats, n-3 PUFA supplements (especially EPA) may be useful in reducing CVD events in moderate-to-high-risk individuals with high triglycerides, and n-6 PUFA may be useful, provided saturated fat is not strongly restricted.^{6,10,15,62,63} Restricting SFA to under 6 % of total daily calories is unsupported; however, a dietary pattern based on healthy fats is recommended.^{6,7,10,40}

Oils are classified on their fatty acid content (which make >95 % of the oil), but the remaining non-glyceride components are also important, and include sterols, alcohols, tocopherols (tocopherols and tocotrienols), phenolic compounds, complex lipids and unique components like ubiquinones, lignans and flavonoids.⁶⁶ Again, data is conflicting: several meta-analyses^{44,45,49,51,53} have reported neutral effects of SFA on CVD outcomes, especially butter and ghee (clarified butter) and even possible benefit of dairy fats in reducing diabetes.^{67–69} The

epidemiological evidence for MUFA is mixed, with mostly neutral, rare negative and some positive results.^{6,10,15,36} The choice is even tougher in India, where the average fat consumption is much lower than in the West (14 % vs 32 % of total calories), the predominant oils are rich in n-6 PUFA (sunflower, safflower, rice bran, proprietary blends), saturated fats are restricted, especially in cities; yet we have growing numbers of individuals with obesity, metabolic syndrome and CVD.^{4,70} This has been attributed to highly refined vegetable oils rich in n-6 PUFA like sunflower oil, which is at odds with guideline recommendations.^{70,71}

Extra-virgin olive oil (EVOO) high in oleic acid (MUFA) and containing phenolic compounds appears to be the healthiest^{10,15,71}; benefits are also reported with canola oil in elevating HDL-C and reducing TGs and BP, as well as preventing recurrent CVD events.^{16,72} The recent practice of blending oils to obtain a healthy fatty acid composition (an optimal mixture of MUFA and PUFA, with other healthy non-glyceride components) may thus be reasonable but is yet to be studied systematically.^{10,72} Coconut oil has been discarded without a fair trial in view of high SFA content: despite having >80 % SFA, lipoprotein changes on a diet enriched with coconut oil may be similar to olive oil.⁷³ This is consistent with decreased TC:HDL-C ratio when lauric acid replaces carbohydrates emphasizing that lipoproteins depend upon foods consumed and cooking methods in addition to fatty acid content.^{6,10,45,73} Considering that these have been staple in many parts of India, coconut and mustard oil (which has similarities with canola oil) deserve further evaluation. Finally, few studies have examined the effects of high temperature cooking on oils: Biochemical properties suggest that SFA are the most heat stable, followed by MUFA, whereas PUFA are unstable and more likely to get denatured with frying.⁴⁵ This could be a reason for the largely neutral results of using ghee,^{67–69} and the better outcomes even in overweight individuals using EVOO^{15,74}; controlled trials studying relative effects of vegetable oils, ghee, mustard and coconut oils are urgently needed.

As of now, the evidence supports the consumption of EVOO and canola oil, followed by healthy proprietary blends that aim for a healthy mix of MUFA, n-3 PUFA and n-6 PUFA (Dietary Guidelines for Indians by National Institute of Nutrition (NIN) recommends a blend of groundnut or sesame or rice bran oil with mustard or canola or soyabean oil). PREDIMED recommended over 4 tablespoons (50 g) of EVOO daily, whereas NIN recommends 20–50 g of visible fat per day depending on activity levels.⁷⁵ Fried foods should be avoided; occasional ($\leq$ once a week) frying may be done in oils with a high smoking point (ghee or clarified butter or mustard oil).

There is inconsistent evidence for recommending specific n-3/n-6 fatty acid ratios of refined oils, as discussed above. As there is data suggesting pro-inflammatory effects of n-6 fatty acids (through arachidonic acid pathway) which reduce the metabolism of dietary alpha linolenic acid to EPA and DHA,⁷⁴ besides purported effects on endocannabinoid system (associated with obesity, inflammation, cancers),^{76,77} but also safety data from a meta-analysis,⁶² it would be prudent to suggest a ratio of <1:5, and definitely <1:10 (commercially available foods typically have ratios over 1:20–60).^{6,10}

4. Proteins

Guidelines advising restriction of meats (and fats) and compensation with carbohydrates have resulted in reduction in the average daily intake of protein to <15 % of total calories, whereas a healthier proportion may be closer to 15–25 %.⁶ Total proteins appear to have little effect on important risk factors like blood sugar, blood pressure, serum lipid levels or obesity and on the basis of limited studies, appear to be neutral for CVD risk.^{6,78} There are isolated reports of some protection from hemorrhagic strokes seen with animal protein intake; a possible protective effect on vascular fragility has been proposed.⁶ Notably, like total fats or carbohydrates, proteins come from a variety of sources, both healthy (nuts, seeds, fish, cheese, curd, etc.) and unhealthy (red meats—processed and unprocessed), suggesting that it is individual

foods and their mode of cooking rather than macronutrients per se that are critical for overall health.^{6,7}

5. Carbohydrates

Carbohydrates account for 50 % or more of the daily calorie intake in most populations. It is difficult to recommend unrestricted consumption of carbohydrates or very low carbohydrate diets (Atkins' diet or Keto diets - these are still shrouded in controversy).^{6,7,26} Whereas carbohydrates are neutral for cardiometabolic health overall, refined and processed carbohydrates are unhealthy and are associated with obesity, diabetes, lipid abnormalities, hypertension and CVD,^{6,10,27–30,79} the effects on inducing/aggravating insulin resistance and weight gain being worse in women, older and sedentary individuals.⁶ The body metabolizes potatoes and refined flour similarly to sugar (as these apparently complex carbohydrates are only glucose molecules strung together), hence, what is important is their preparation—the fiber content, level of processing and whole grain content, all of which determine the glycemic response to a food (determined by the glycemic index (GI) and the glycemic loads of foods).^{6,7,24,27–30}

6. Wheat and wheat products

The Green Revolution transformed agriculture by hybridization of the wheat crop: This grew faster, was winter hardy and gave a larger harvest; however, the protein gluten was different, and Amylopectin A provided high glycemic index (white bread has GI similar to sugar).⁸⁰ Current wheat (*Triticum aestivum*) has 70 % carbohydrate, 10–15 % each of indigestible fiber and proteins and a small amount of fat.^{6,80} The glycemic response to grains is low if eaten whole or with minimal processing (quinoa, brown rice or steel-cut oats—the bran protects the starchy endosperm portion from easy digestion), is higher if the grain is milled (bran is present, but the starchy endosperm is more exposed leading to early digestion, e.g., chapati made from traditional flour, whole-grain breads, some cereals) and is highest if the bran and germ are totally removed by processing (leaving the easily digestible starchy portion e.g., white bread, white rice, most cereals like corn flakes and highly refined flour), again reinforcing foods over macronutrient composition.^{6,7}

As wheat products are ubiquitous, it is reasonable to continue moderate intakes (up to 3 servings a day) including chapatis, breads, or cereal.

7. Eggs

It is now accepted that dietary cholesterol has neutral associations with CVD, and in usually consumed amounts, has minor effects on serum cholesterol.^{81,82}

Eggs, a major source of cholesterol, are also rich in complete proteins and fat-soluble vitamins.^{81,82} Large observational studies suggest no association of eggs with CVD, rather a possible association with diabetes in people consuming >14 eggs per week, which is controversial as all possible results (negative, neutral and positive associations with diabetes) are reported.^{83–86} Three large studies on egg consumption published recently reported contrasting results. An American study⁸⁴ of over 29,000 adults showed a higher risk of CVD and all-cause mortality, whereas a Chinese study⁸⁵ in half a million adults suggested lower risk of CVD with ≤ 1 egg/day. PURE found no association of egg intake (≥ 7 /week vs < 1 egg/week) with lipoprotein levels, major CVD events or mortality in 1,77,000 subjects.⁸⁶

In summary, eggs would appear to be healthy if replacing foods like red meats, fried foods, and refined carbohydrates, but may be less healthy than non-starchy vegetables, fruits, and fish, making 1 whole egg a day acceptable in healthy individuals (two in those who are primarily vegetarians).^{6,82}

8. Milk and milk products

Conventionally lumped together as a single entity, usual dietary recommendations are to use low-fat dairy to reduce saturated fats, even though low-fat milk in children (but not adults) is associated with more weight gain.⁶ Like all foods, milk products have heterogeneous effects on the metabolism; for instance, curd and cheese, but not milk, are associated with a reduced risk of diabetes and weight loss if replacing carbohydrates; this suggests a beneficial effect of fermentation on milk, possibly because of the synthesis of vitamin K2 by bacterial cultures or the presence of probiotics in curd.^{6,10,67–69} Several meta-analyses report protective effects of dairy fat and butter on diabetes risk, due to the zero GI, and no association with CHD, stroke or total mortality.^{67–69} The available evidence points to the benefits of regular consumption of curd and probably cheese as well^{6,10,67–69}; three recent analyses report little difference between low-fat and full-fat varieties in 3 US and 1 Italian cohort, further strengthening the case for recommending dairy products to healthy children and adults.⁸⁷

Healthy children and adults may consume low-fat or full-fat milk as per personal choice, whereas dairy fat and butter appear to be metabolically safe with possible protective effects on diabetes risk. The recommendation is to advise consumption of curd and cheese over milk for weight loss and reducing the risk of diabetes.

9. Coffee and tea

Coffee is associated with reduction in the risk of diabetes in a dose-dependent manner, whether with caffeine or decaffeinated. The effect on CHD and stroke is more complex: observational studies appear to suggest a J-shaped relation, i.e., protective till 3–4 cups per day and possibility of harm above 5–6 cups per day; randomized trials have been small and inconsistent.⁸⁸ A large study recently reported lower CVD and mortality with coffee consumption with maximum protection at 2–3 cups a day.⁸⁹ Tea is associated with similar benefit in reducing the incidence of CVD and diabetes, especially with >3–4 cups per day. In sum, the recommendation is to permit the consumption of tea and coffee, with some evidence of a benefit of black and green tea in reducing CVD risk.⁹⁰

10. Sodium

Sodium increases blood pressure in a dose-dependent fashion, especially with intakes >5 g/day and in individuals with hypertension.^{91,92} One meta-analysis of prospective cohort studies reported an increased risk of stroke and stroke mortality at high intake,⁹² however, there is limited evidence of a linear association with cardiac or total mortality in RCTs or cohort studies.^{91,93–100} Rather, several meta-analyses of prospective observational studies report a J/U-shaped mortality curve, suggesting a troubling increase in all-cause mortality with low-sodium diets both in hypertensive as well as normotensive individuals.^{93,96} The average global consumption of sodium is estimated at 3.95 g/day (approximately 10 g of salt)¹⁰¹ and exceeds most dietary guidelines that recommend <2–2.3 g/day, whereas the number of people consuming this level is <10 % in US and even lesser elsewhere (of over one lakh individuals in PURE, sodium excretion was <3 g in 11 % and <2.3 g in just 0.2 %).^{95–97,100} A recent study found no benefits in restricting sodium <2.3 g/day in heart failure, making even this indication suspect.¹⁰²

Guidelines are based on one 30-day diet trial¹⁰³ showing BP lowering and short-term safety of 1.5 g sodium, and two longer trials of intensive dietary counselling.^{104,105} The bigger of these¹⁰⁴ managed to achieve an intake of 3.2 g/day in the intervention arm (<10 % achieving <2.3 g) compared to 4 g in the control arm. While some investigators claim potential 32 % reduction of CVD risk with restriction of daily sodium intake to 2.3 g,⁹¹ this remains controversial as this target has not been achieved anywhere outside a trial. Other problems include difficulty in

reliable measurement of sodium intake, and in achieving or quantifying such severe restriction (maximum sodium decrease in populations has been 0.5 g/day with intensive counselling and food industry involvement).^{95,98}

Two recent reports rekindled interest in salt restriction: an observational study reported an association of added salt on table with mortality¹⁰⁶ and a trial demonstrated reduction in CVD and mortality in high-risk hypertensives using salt substitutes.¹⁰⁷ Of note, the former analysis reconfirmed that the harms of added salt were neutralized in those consuming potassium-rich foods like fruits, vegetables and nuts.^{95,106} This is very relevant in India with estimated intake of 3.72–5 g/day^{101,108} where >80 % of the salt is added (in comparison to 5–20 % in high income countries) at the table.¹⁰⁸

As there is clear benefit only for hypertensive individuals with high intakes, and possible risks for low intakes in all, a practical approach would be to advise restriction of added salt by educating populations and industry in order to achieve the WHO 2025 goal of reducing population salt intake by 30 %.^{95,98,100,106,107}

11. Miscellaneous

Compounds like flavonols (onions, broccoli, many fruits, especially berries, tea), flavones (parsley), isoflavones (soy), catechins (cocoa, grapes, red wine, tea), etc. have documented cardiometabolic effects, though their variability precludes definite guidelines; several food items deemed healthy (EVOO, nuts and berries) are rich in these compounds.^{6,10,15,71}

Though observational studies suggested a reduced risk of CVD with vitamins, RCTs involving folate, vitamin B group, beta-carotene, and vitamins C and E have been neutral.⁶ Reasons could be that the duration of treatment was not long enough, or that these are more effective for primary than secondary prevention. It is also possible that the benefits seen with vegetables, fruits and nuts may be secondary to factors (dietary or non-dietary) other than vitamins, hence their benefit is not replicated by synthetic vitamins. Vitamin D has invited extensive investigation recently: there is currently no evidence for its role in preventing CVD.⁶ Higher intake of potassium may be useful, e.g., fruits, vegetables and nuts like the Mediterranean diet or the modified DASH diet (with higher fat) lower the risk of CVD, and are very beneficial, if taken regularly.^{6,10,15,17,18,95,96,98,100}

12. Dietary approach to lipid management

As LDL-C is widely recognized as causative for ASCVD, the focus of clinical management is on lowering LDL-C using statins and other agents.^{1,2} However, from a community health perspective, it's important to recognize two different types of lipid disorders: isolated high LDL-C and metabolic/atherogenic dyslipidemia, as these differ both in etiology as well as dietary management. Isolated LDL-C elevation is less common and responds best to restriction of trans as well as saturated fats, whereas the more common metabolic dyslipidemia (high triglycerides, low HDL-C) is managed better by restriction of total calories, especially sugars and refined carbohydrates.^{6,31,36,37,40,109}

In view of lack of robust evidence linking specific nutrient consumption with optimal health and the difficulties involved in calculating such figures, a practical approach is presented. Also refer to [Tables 1–3](#) for additional information.

12.1. Category 1: foods for which there is consensus that they are unhealthy

These include trans fats, processed meats, some oils (hydrogenated vegetable oils), refined carbohydrates, starches, sugars, sugar sweetened beverages (SSB) and excess salt. These should be avoided, except for fresh unprocessed meats, which may be consumed occasionally (discussed below).

Table 1
Healthy or health neutral foods with evidence base.

Broad consensus	General consensus, few doubts	Controversy as to evidence and prevailing guidelines
Fresh vegetables, dietary fibre	n-6 PUFA	Low fat milk
Curd/yogurt	Plant derived MUFA	Cheese, dairy fat
Fresh fruits	Certain oils (EVOO, canola)	Certain oils (sunflower, safflower)
Nuts and seeds	White meats	Eggs
Whole grains	Phenolics, flavones	Probiotics
Fish and fish oil	Tea/coffee	Vitamin D
Legumes n-3 PUFA		

EVOO: Extra-virgin olive oil, PUFA: Polyunsaturated fatty acids, MUFA: Monounsaturated fatty acids.

Table 2
Foods considered unhealthy.

Broad consensus	General consensus, few doubts	Controversy as to evidence and prevailing guidelines
Processed meats	White potatoes	Saturated fats, butter Coconut oil Dietary cholesterol
Partially hydrogenated vegetable oils (TFA)	Unrestricted unprocessed red meats	
Refined grains and starch Sugars and SSBs Excess sodium		Whole fat milk Non-calorie sweeteners Moderate amounts of unprocessed meats
Excess alcohol Sweets and desserts		

TFA: Trans fatty acids, SSBs: Sugar sweetened beverages.

Industrially produced trans-fatty acids (TFA) in hydrogenated oils (developed to increase the shelf life of oils without becoming rancid, and making them more suitable for frying, baking and packaging foods) raise LDL-C, TGs and Apo B, while lowering HDL-C and Apo A; other effects include insulin resistance, visceral obesity, inflammation, damage to vascular endothelium and possibly increased cardiac arrhythmias.^{6,10,11} As frying at high temperatures generates further TFA, frying in the same oil repeatedly, a common practice in sweet shops (and most Indian homes) should be avoided. Sugars should be restricted, being worse than “empty calories”; SSBs are highly addictive, being consistently linked with diabetes, heart disease and cancers.^{41,42,110–114}

12.2. Category 2: foods about which there is consensus that they are useful or neutral for CVD

Compelling evidence exists for the benefits of eating non-starchy vegetables, especially green, leafy vegetables,^{3,4,6,7,115} a moderate consumption of fruits,^{3,4,6,7,115} nuts,^{6,10,116} seeds, fatty fish and fish oils.^{6,7,10,11} Other healthy foods include whole grains, legumes,¹¹⁶ some extra-virgin (unrefined) oils and non-fat components in vegetable oils like phenolics, whereas some dairy items (curd and cheese), white meats, tea and coffee are neutral or possibly healthy. These food items consistently improve metabolic and biochemical parameters, are neutral or beneficial for weight maintenance and reduce CVD risk; low intake of whole grains and fruits were among the three leading dietary risk factors for death globally.³ These foods can be consumed liberally, especially if they replace foods in category 1.

12.3. Category 3: Controversial foods

Saturated fats, wheat products, meats, other dairy items (low-fat or full-fat milk, butter, ghee), eggs, vitamin D, specific oils and fatty acids, and probiotics have conflicting data; there is too little evidence to draw conclusions upon some other popular recommendations like organic

Table 3
General dietary recommendations.

Foods	Recommended amount (servings)	Size of 1 serving
Fresh vegetables, legumes	3–5 per day	1 cup raw veg, 1/2 cup cooked or cut veg, veg soup
Fresh fruits	2–3 per day	1 medium size fruit, 1/2 cup fresh or unsweetened canned fruits, one piece about the size of your palm
Nuts and seeds	1 per day	1 ounce or a small handful of nuts
Whole grains	2–3 per day	1 medium chapati (multigrain/ flour with bran, 1 slice whole grain bread, 1 cup high fibre cereal, 1/2 cup cooked brown rice
Dairy	2–3 per day	1 cup milk/curd (preferred), 1 ounce (~28 g) cheese
Fish/seafoods	2–3 per week	100 g
Vegetable oils	2–5 per day	1 tsp oil, prefer unrefined virgin/ extra virgin oils
Processed meats	Maximum 1 per week	50 g
Unprocessed meats	1–3 per week	100 g
Refined grains	<1–2 per day	1 medium chapati (refined flour), 1 slice bread, 1/2 cup rice or cereal, 1 small sweet or dessert
Hydrogenated oils	Avoid as much as possible	Foods made with partially hydrogenated oils, fried foods (especially re-frying in same oil)
Sugar sweetened beverages	Avoid as much as possible	300 ml
Sodium	<2.3 g recommended, 2–5 g may be acceptable per day	Not relevant
Eggs	1–2 whole eggs per day	1 egg
Alcohol	1–2 small drinks males 1 small drink females a few times a week	10 g of ethyl alcohol
Tea	3–5 cups per day	2 g loose tea in 200 ml or 8 oz boiled water
Coffee	1–3 cups per day	1 tablespoon full of coffee in 6 oz (~170 ml) cup

foods, grass-fed or grain-fed, modified or non-genetically modified foods. Meats are controversial as there are opposing recommendations (fresh vs processed, white vs red meats); presently, it is reasonable to recommend intake of fresh unprocessed meats (< once daily, up to 1–3 times a week) as the evidence implicating them in causing diabetes or CVD is weak.^{117–121} There is little data to support the recommendation that meats should be chosen based on their fat content^{6,10}; a recent nutritional guideline has rather given a weak recommendation to continue current levels of meat intake due to the very poor evidence linking meats to CVD, diabetes or cancers.⁵⁸

Several questions remain unanswered. These include, but are not limited to:

1. Confirmation of the benefits of consuming PUFA instead of other fats and carbohydrates.
2. The ideal cooking oil and the optimal n-3/n-6 ratio.
3. The putative health effects of low carbohydrate diets (<26 % daily calories)

13. Conclusion

Carbohydrates, fats and proteins are not harmful in themselves: Food items and their preparations are healthy or unhealthy; this knowledge directs us to choose healthy foods rather than macronutrients. The available data supports the consumption of a plant-based omnivorous diet, rich in natural foods with low consumption of salt and meats, and minimal intake of sugars and processed foods. Fats are not to be treated as poison, and nor are carbohydrates; rather a balanced diet contains all

of them. Perhaps a reduction of carbohydrates by 10–15 % (especially sugars, SSBs and refined carbohydrates) and adding some healthy fats and proteins is the best option.^{6,10} Furthermore, incomplete data from genetically distinct populations consuming different foods using dissimilar cooking methods cannot be applied blindly in our country, and we need more local studies to clearly define the best foods and cooking practices in our context.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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